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**Title: Crypto Taxation: “Challenges in Defining and Regulating Digital Assets
in the EU and Beyond”**

THESIS



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Abstract

Over the past decade, digital assets specifically cryptocurrencies have evolved from niche technological experiments into major components of global financial markets. Their speedy expansion presents unmatched challenges for legal systems like never before, especially in the field of taxation. Traditional tax frameworks were not calibrated to address decentralised, pseudonymous, and borderless digital assets, resulting in conceptual and practical fickleness.

While several jurisdictions, including EU Member States, have initiated targeted regulatory and tax measures, significant uncertainty remains with regards to the legal classification of crypto assets, the identification of taxable events, valuation mechanisms, and enforcement capabilities. The increasing alignment between EU initiatives such as the Markets in Crypto-Assets Regulation and DAC8 and international frameworks originated by the OECD, particularly the Crypto-Asset Reporting Framework, highlights the importance and need for a comprehensive legal analysis.

This thesis seeks to examine the challenges associated with defining and taxing digital assets, assess the adequacy of existing regulatory responses, and explore principled approaches for evolving clearer, fairer, and more exercisable crypto tax regimes.

Table of Abbreviations

Abbreviation	Full Form
AEOI	Automatic Exchange of Information
AML	Anti Money Laundering
ART	Asset Referenced Token (Under MiCA)
BFH	Bundesfinanzhof (Federal Fiscal Court of Germany)
BMF	Bundesministerium der Finanzen (German Federal Ministry of Finance)
BNC	Bénéfices Non Commerciaux (Non-commercial Profits)
BOFIP	Bulletin Officiel des Finances Publiques – Impôts (Tax Bulletin for French Tax Administration)
CARF	Crypto-Asset Reporting Framework (OECD)
CASP	Crypto-Asset Service Provider
CBDC	Central Bank Digital Currency
CFTC	Commodity Futures Trading Commission (US)
CJEU	Court of Justice of the European Union
CRS	Common Reporting Standard
DAC8	Eighth Directive on Administrative Cooperation (Directive (EU) 2023/2226)
DeFi	Decentralised Finance
EBA	European Banking Authority
EMT	E-Money Token (under MiCA)
ESMA	European Securities and Markets Authority
EStG	Einkommensteuergesetz (Income Tax Act, Germany)
FATF	Financial Action Task Force
GDPR	General Data Protection Regulation (Regulation (EU) 2016/679)

HMRC	His Majesty's Revenue and Customs (UK)
IRS	Internal Revenue Service (US)
KYC	Know Your Customer
MAS	Monetary Authority of Singapore
MiCA	Markets in Crypto-Assets Regulation (Regulation (EU) 2023/1114)
NFT	Non-Fungible Token
OECD	Organisation for Economic Co-operation and Development
RCASPR	Reporting Crypto-Asset Service Provider (under DAC8/CARF)
SEC	Securities and Exchange Commission (US)
TFEU	Treaty on the Functioning of the European Union
TFR	Transfer of Funds Regulation (Regulation (EU) 2023/1113)
VARA	Virtual Assets Regulatory Authority (Dubai)
VAT	Value Added Tax

Chapter 1: Introduction and Methodological Framework

1.1 Introduction and Background

The fast-paced digitization of the global economy has primarily challenged traditional principles of taxation. Over the last era, crypto assets have developed from a niche technological experiment into a unique asset class with a market capitalization that has started challenging established financial sectors and has started to look like a strong contender. Yet, despite this economic maturation, tax laws have struggled to keep pace. Traditional tax systems are predicated on the concepts of territoriality, identifiable intermediaries, and sovereign currency; three features that decentralised digital assets explicitly seek to dodge.¹

As of 2025, the European Union has developed a thorough regulatory baseline through the Markets in Crypto-Assets Regulation (MiCA). However, while financial regulation has regularized, tax treatment remains disintegrated. This thesis theorizes that the divergence between “*regulatory*” definitions (designed for market stability) and *tax* definitions (designed for revenue collection) creates a “legal limbo” that facilitates tax evasion, complicates compliance, and distorts the Internal Market.

The challenge posed is significant, it is not only about classification, but of enforcement. Crypto assets operate on “pseudonymous ledgers” meaning by that they operate using false or fictitious name, often without a central counterparty. This feature, while innovative, wears away the informational visibility that tax authorities rely upon. Subsequently, the intersection of tax enforcement and AML frameworks have become the critical boundary for revenue authorities. As nations are moving towards implementing the OECD’s CARF and the EU’s Eighth Directive on Administrative Cooperation, the central question remains, “can a territorially bound tax system effectively capture a borderless, decentralized value transfer system?”²

1.2 Research Problem

The heart of the thesis and the central issue addressed in this thesis is the absence of a uniform, tax-specific legal taxonomy for digital assets, which has ended up in legal uncertainty and enforcement gaps. While MiCA has provided a “passportable” definition for crypto assets for financial services, Member States retain sovereignty and control over direct taxation. This has led to a landscape where a single token may be treated as a “commodity” in one jurisdiction, “private money” in another, and an “intangible asset” in a third.

This definitional uncertainty gives rise to two distinct problems:

¹ OECD, ‘Taxing Virtual Currencies: An Overview of Tax Treatments and Emerging Tax Policy Issues’ (OECD 2020)

² Council Directive (EU) 2023/2226 of 17 October 2023 amending Directive 2011/16/EU on administrative cooperation in the field of taxation [2023] OJ L 2023/2226 (DAC8).

- a) Firstly, for compliant taxpayers and businesses, it increases the administrative burden and risk of double taxation, potentially impeding the competitiveness of the EU digital economy.
- b) Secondly, for tax authorities, the lack of certainty and clarity combined with the “pseudonymity” of blockchain transactions creates significant opportunities for base erosion and non-compliance. Although AML regulations do serve as a partial bridge by requiring intermediaries to identify users, the growing influence of DeFi and unhosted wallets threatens to evade these checkpoints entirely.³

Furthermore, the "Beyond" element of this research points out that the EU does not operate in a vacuum. As the US and UK are developing their own competing tax and reporting frameworks, the EU's ability to enforce its tax claims without driving capital offshore is tested. The research problem is therefore how to design a tax framework that is robust enough to prevent evasion but clear enough to foster innovation. Moreover, this thesis is aimed to cope up with the legal developments especially in this era where technology fast outpaces the legal developments and policy making, it is an attempt to not only catch up but to remain consistently up to the mark when it comes to legal policy making and implementation to minimize the grey areas and keep the innovation regulated and monitored in order to prevent financial crimes and so on.

1.3 Research Questions

To address these challenges, this thesis comprises of the following three research questions:

- 1. How does the lack of a uniform legal definition of crypto assets create challenges for their taxation?**
- 2. What enforcement and compliance challenges do crypto-assets pose for tax authorities, particularly in relation to anonymity, cross-border transactions, and money laundering risks?**
- 3. To what extent do existing European Union and OECD regulatory frameworks address the tax-related challenges arising from crypto-assets?**

1.4 Scope

The scope of this thesis is mainly shedding light on the legal and administrative aspects of taxation. It centers on Direct Taxation (Income and Capital Gains) and the administrative cooperation required to enforce it. Discussion on VAT is present only where relevant to the definition of assets (following the jurisprudence).

Jurisdictionally, fundamental focus is on the European Union. However, to address the "Beyond" aspect of this research topic, comparative analysis is drawn from the US and UK. These jurisdictions are chosen firstly, as they represent the dominant alternative legal system “Common Law” and secondly, they compete directly with the EU for crypto capital.

³ Financial Action Task Force (FATF), *Updated Guidance for a Risk-Based Approach to Virtual Assets and Virtual Asset Service Providers* (FATF 2021) para 45.

Regarding Money Laundering, this thesis does not seek to provide a comprehensive analysis of financial crime. Instead, it examines AML strictly through the perspective of tax enforcement specifically, how AML due diligence (CDD/KYC) serves as the primary information-gathering and processing tool for tax authorities.⁴

1.5 Methodology

1. **Doctrinal Analysis:** The research analyses primary EU legislation (MiCA, DAC8), CJEU case law, and national tax guidelines. It evaluates the legal text against the principles of legal certainty, tax neutrality, and enforceability.
2. **Comparative Analysis:** The thesis contrasts the civil law approach of the EU (codified regulation) with the common law/regulatory enforcement approach of the US (IRS rulings and court cases) and the UK (HMRC guidance).
3. **Contextual Timeline:** The analysis is conducted as of December 15, 2025. It assumes MiCA is fully applicable and evaluates DAC8 at the moment of its transposition deadline, offering a contemporary critique of implementation challenges.

1.6 Roadmap

The Tension Between Digital Assets and Analog Law

The primary argument that the central thesis explores a fundamental structural crisis in the modern global economy, the collision between “borderless, fluid digital assets and territorially anchored, rigid tax laws”. It posits that while the European Union has successfully harmonised the commercial identity of digital assets through MiCA, a critical gap remains regarding their “fiscal” identity. This research investigates whether a "Single Market" for crypto assets can realistically exist or not, when the same token transforms its legal identity from currency to commodity to property simply by crossing a Member State border.

1. The "Panopticon" Trap (Enforcement) A key area of inquiry is the EU’s reliance on a "Panopticon" model of enforcement (DAC8), which instructs total visibility over centralised intermediaries. The thesis examines whether this strategy is sufficient in an era of unhosted wallets and DeFi. It questions if the current regulatory focus on "on-ramps" inadvertently creates a regressive system where compliant businesses must endure the weight of mass reporting, while the peer-to-peer economy drifts further into cloudiness.

2. The Valuation Crisis (Business Law) Beyond classification, the research highlights a practical crisis in “valuation” that threatens the viability of crypto-business models. Tax laws generally assume assets have a clear "Fair Market Value" in fiat currency at the moment of transaction. The thesis investigates the "Valuation Nightmare" of applying these principles to high-frequency DeFi transactions and illiquid assets. It questions upon the fact that can traditional "capital gains"

⁴ Peter Ceron and A. Yanev, ‘The intersection of AML and Tax Evasion in the Crypto-sphere’ (2022) 30(2) *Journal of Financial Regulation and Compliance* 155.

realization events function in an ecosystem of continuous, algorithmic token swaps, or is a fundamental rethink of the "taxable event" required?

3. The Global Competitive Divide (International Law) Lastly, the research frames the international landscape not merely as a race for tax revenue generation, but as a deeper divergence in legal philosophy:

- **EU (Civil Law):** Investigated for its focus on codification and transparency, prioritizing and enhancing consumer safety and market integrity.
- **UK (Common Law):** Analysed for its tilt towards property rights and flexibility, particularly through the “*Property (Digital Assets etc) Act 2025*”, and how this impacts institutional certainty.
- **US (Enforcement):** Examined as a case study in litigation-driven regulation, offering a counter-narrative to the European approach to enhance and streamline globalized taxonomy.

The pertinent and judicious challenge and the ultimate question posed by this research is whether the "Definitional Quagmire" highlighted in Chapter 2 renders the surveillance mechanisms of Chapter 4 as “legally fragile.” This research induces the readers to consider whether the “EU’s current trajectory superimposing analogue reporting rules onto digital protocols is a sustainable long-term solution, or if it merely sets the stage for a decade of constitutional and administrative conflict.” The final chapters ponder upon these competing legal realities to determine if a unified "Tax Taxonomy" is the missing piece of the “Digital Single Market” puzzle.

CHAPTER 2: The Definitional Quagmire: Tax Law vs. Financial Regulation

2.1 Introduction: The Classification Void

One of the principal trials in taxing digital assets lies in their *chameleon-like* nature. A single crypto-asset can function at the same time as a means of payment, a store of value, a representation of capital in a business, or even a utility access right. However, traditional tax structures and systems are built on rigid, mutually exclusive categorizations, an asset is either “currency, commodity, property, or security.” The “Definitional Quagmire” is termed to give a perspective of the attempt of forcing these decentralised digital assets into these pre-digital legal buckets which is a state of legal incoherence where the tax treatment of an asset depends entirely on the jurisdiction in which the taxpayer resides rather than the economic element of the asset itself.

This chapter delves into the divergence between the regulatory definitions established by the European Union’s Markets in Crypto-Assets Regulation and the considerably uneven tax definitions applied by national authorities. It debates on pertinent points that while “MiCA” has successfully streamlined the prudential classification of crypto-assets to make matters easier for the internal market, this harmonisation has not extended to taxation specifically. Therefore, businesses functioning across borders face an extra burden. The burden that they must comply with a unified EU regulatory framework while navigating the harshness of the national tax regimes.

2.2 European Union’s Attitude: Regulatory Unity & Fiscal Fragmentation

The European Union’s methodology to crypto assets can be determined by a sharp dichotomy between indirect taxation (VAT), which has been streamlined and harmonised via the Court of Justice of the European Union, and direct taxation, which remains under the sole competence of Member States.

2.2.1 Case Law of Skatteverket v Hedqvist: The foundational case law for crypto taxation in the EU is *Skatteverket v Hedqvist* (2015). In this landmark judgment, the CJEU was asked to determine whether the exchange of Bitcoin for fiat currency constituted a supply of services for VAT purposes. The Court ruled that “Bitcoin, having no other purpose than to serve as a means of payment, must be treated as currency, bank notes and coins used as legal tender under Article 135(1)(e) of the VAT Directive.”⁵

While this ruling is essential and it laid down crucial certainty for the VAT treatment of payment tokens effectively exempting them from VAT and preventing the double taxation of the currency itself, its reach is strictly restricted towards VAT treatment. The court’s reasoning did not extend to direct taxation rather was only confined towards the interpretation of “currency, bank notes and coins” for VAT exemption purposes and did not extend to direct taxation, which remains within the scope of Member States under Articles 110–113 TFEU. However, *Hedqvist* illustrates the

⁵ Case C-264/14 *Skatteverket v David Hedqvist* [2015] ECLI:EU:C:2015:718, paras 55–57.

systemic difficulty of applying analogue concepts to digital assets. While the CJEU functionally treated Bitcoin as currency to ensure VAT neutrality, national tax authorities dealing with income tax were left without EU-level guidance and ruling. Therefore, Member States have relied and trusted on their sovereign discretion to classify crypto-assets, leading to the current fragmentation where the same asset is VAT-exempt across the EU but subject to differing income tax treatments nationally.

2.2.2 MiCA Detachment, With the full application of the MiCA Regulations in late 2024, the EU brought up and introduced a robust legal taxonomy for digital assets. Before discussing on MiCA it is pertinent to know that MiCA explicitly excludes taxation from its scope, respecting the principal of fiscal sovereignty. Therefore, we cannot criticize this regulation for failing to harmonize the tax bases as it doesn't have the mandate to do so. Moving Forward, MiCA divides the market into three main categories which are referred herein below:

- **Asset-Referenced Tokens:** Tokens purporting to maintain a stable value by referencing a basket of currencies or assets.
- **E-Money Tokens:** Tokens referencing a single official currency.
- **Other Crypto-Assets:** A catch-all category for utility tokens and unbacked cryptocurrencies like Bitcoin.⁶

In light of a business law perspective, MiCA gives legal certainty for “issuers and service providers (CASPs).” However, this regulatory certainty does not mean that there is tax certainty as well. There is a significant *de lege ferenda* challenge. There remains a friction when a harmonised regulatory layer (MiCA) interacts with a non-harmonised tax layer. While a token may be passported across the EU as a uniform “E-Money Token” for regulatory purposes, its tax treatment varies wildly. This thesis argues not that MiCA failed, but that the absence of a complementary tax directive and uniformity with DAC8 and member states in terms of definitions establishes a dichotomy that practically hinders the seamless operation of the Internal Market, even if such diversity is legally tolerated under current EU law.

2.2.3 National Discrepancy: The German vs. French Model The practicality and the consequences of this lack of harmonisation is evident in the contrasting approaches of the EU's two largest economies which I laid down below and touched above:

- **Germany (The 'Private Money' Approach):** German tax law “(Einkommensteuergesetz)” sees crypto assets not as financial items / products but as “other economic goods”. Crucially, under “Section 23 EStG, gains from crypto assets held for more than one year are distinctively tax-exempt for private individuals.” However, this treatment is not uniform, assets held as business property are subject to trade tax and lack this exemption. Furthermore, administrative guidance “BMF-Schreiben” has clarified that distinct activities, such as mining or staking, may trigger different tax events compared to simple holding, though 'edge cases' regarding the taxation of liquidity pool tokens remain

⁶ Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets (MiCA) [2023] OJ L 150/40, art 3(1).

subject to debate. This establishes a very strong incentive for long-term holding ("HODLing") but complicates high-frequency trading.⁷

- **France (The 'Intangible Asset' Approach):** Whereas in a completely opposite direction, France treats digital assets as a specific category of “movable property”. Since the PACTE Law and subsequent Finance Acts, gains are subject to a straight “Flat Tax (Prélèvement Forfaitaire Unique) of 30%.” Completely opposite to Germany, there is no tax-free exit after one year, but crypto-to-crypto trades are always tax-neutral, delaying the taxable event until a person converts it to fiat currency.⁸ While France offers a trailblazing “crypto-to-crypto” exemption deferring taxation until conversion to fiat this neutrality has interpretive limits. Administrative doctrine (BOFIP) suggests that high-frequency trading may still be reclassified as professional activity subject to progressive income tax rates (BNC), creating a layer of uncertainty for active traders despite the statutory clarity."
- For an international business, this divergence is very crucial. A portfolio manager offering their services to both German and French clients must build different tax-reporting engines for each, accounting for the "1-year speculation period" in Germany and the "fiat-conversion trigger" in France.

2.3 The 'Beyond' Perspective & Importance

To fully understand the global competitive environment, it is important to compare and contrast the EU's approach with the Common Law jurisdictions of the United States and the United Kingdom, which are dominating the global crypto-financial market.

2.3.1 United States: The United States obeys strictly to a "Property" classification, as established in IRS “Notice 2014-21.” This notice provided clarification that "convertible virtual currency is treated as property" and that the general tax principles which are applicable to property transactions will apply.⁹

While we can say that this gives / provides "certainty" in a strict legal sense, it also gives rise to risky and extreme administrative friction for the use of crypto as a medium of exchange. Every purchase of goods or services using crypto sparks a capital gains realization event. For example, like purchasing a cup of coffee with Bitcoin requires the taxpayer to calculate the difference between the Bitcoin's cost basis and its “FMV” at the same moment of the transaction. Similarly, the US approach to "Hard Forks" where a blockchain splits into two is also very aggressive. Under “Revenue Ruling 2019-24,” the IRS pushes that a taxpayer receives taxable income from a hard fork (like an airdrop) the moment they have "dominion and control" over the new token often

⁷ Einkommensteuergesetz (Income Tax Act) (Germany) s 23(1) No 2. Bundesministerium der Finanzen (BMF), ‘Einzelfragen zur ertragsteuerrechtlichen Behandlung von virtuellen Währungen und von sonstigen Token’ (10 May 2022) IV C 1 - S 2256/19/10003 :001.

⁸ Code général des impôts (General Tax Code) (France) art 150 VH bis: *Loi n° 2019-486 du 22 mai 2019 relative à la croissance et à la transformation des entreprises* (PACTE Law).

⁹ Internal Revenue Service (IRS), ‘Notice 2014-21: Virtual Currency Guidance’ (2014) IRB 2014-16

creating a tax liability for an asset the taxpayer never asked for and may not be able to sell immediately.¹⁰

2.3.2 The United Kingdom: UK's approach, steered by HMRC, is by far in my opinion the most flexible of the major jurisdictions. HMRC's "Cryptoassets Manual" explicitly says that crypto assets are "not currency or money," but it refuses to shoehorn them into a single definition. Instead, the UK operates and applies a functional test based on the "Badges of Trade."¹¹

HMRC makes a distinction between three types of tokens for tax purposes:

- **Exchange Tokens:** Intended to be used as a method of payment (e.g., Bitcoin).
- **Utility Tokens:** Providing access to goods or services.
- **Security Tokens:** Providing rights or interests in a business (similar to shares).

What's more important is that for businesses, the UK creates a distinction between "investing" (subject to Capital Gains Tax) and "trading" (subject to Income Tax). The overall threshold for "trading" is high, meaning most individual investors are falling under the more favorable Capital Gains regime (now 24% for higher rates as of late 2025 adjustments, subject to allowances). This approach is more practical and functional. Moreover, it allows the tax treatment to evolve with the technology, avoiding the harsh and rigid "property trap" of the US while offering more clarity than the fractured and disintegrated EU structure.

2.4 Effect on Business and the Internal Market

The absence of a harmonised tax definition that aligns with MiCA creates three critical issues / problems in the European Internal Market:

- **Compliance Costs:** A Crypto-Asset Service Provider delivering its services on a cross-border scale under MiCA must maintain 27 different tax logic systems to generate compliant reports for users in different Member States.
- **Location Problems:** Businesses are incentivized to locate in Member States with favorable definitions (e.g., those recognizing "staking" rewards as capital growth rather than immediate income), rather than where they are most economically productive.
- **Risk Legally:** Unpredictability around complex DeFi transactions (e.g., liquidity pool tokens) leaves businesses exposed and open to "retroactive" tax reclassifications.

¹⁰ Internal Revenue Service (IRS), 'Revenue Ruling 2019-24: Virtual Currency Hard Forks and Airdrops' (2019)

¹¹ HM Revenue & Customs (HMRC), 'Cryptoassets Manual' (HMRC, updated November 2025)

2.5 Insights

The "Definitional Quagmire" remains as the tax law is inherently backward-looking, relying on case laws and statutes designed for purely the physical economy. While it is a partial win that the EU has efficaciously modernized its financial regulation through MiCA, the tax framework remains an outdated patchwork. The result? A paradox, a digital asset may be a "regulated financial instrument" for the purposes of selling it, but a "private commodity" for the purposes of taxing it. In the upcoming chapters, it will be highlighted that this lack of definitional clarity not only hampers and disturbs businesses but also creates the perfect conditions for the "Enforcement Gap" where anonymity and confusion both combine to assist and enable widespread non-compliance.

CHAPTER 3: The Enforcement Gap: Anonymity, AML, and the Limits of Territorial Taxation

3.1 The Liability & Enforceability Puzzle

A principle saying of tax law is that tax liability arises from economic substance. Let's say, if an individual generates a capital gain, a tax debt is created, irrespective of the medium of exchange. Critically, the growing rise and influence of crypto assets has exposed a significant weakness in modern tax administration, the widening gap between "legal liability and practical enforceability."

This chapter delves into the "Enforcement Gap" the significant difference between the theoretical tax obligations of crypto-investors and the ability of tax authorities to detect and verify non-compliance. It debates upon the fact that while the EU has successfully deployed AML frameworks and laws to shed light on the "on-ramps" and "off-ramps" of the crypto-economy "(via centralised exchanges)", the heart of the ecosystem, DeFi and unhosted wallets remains largely a grey area. As of late 2025, tax authorities find themselves in a paradox and although they have more legal tools than ever before, yet the target of their enforcement is becoming increasingly obscure.

3.2 Technical Obstacles: Anonymity vs. Pseudonymity

In order to study the enforcement challenge, we must be able to create a distinction between the "public nature of the blockchain and the private nature of identity". What this means is that most blockchains are pseudonymous, every transaction is visible, but the actors are represented by alphanumeric strings (public keys).

For a tax authority, a blockchain ledger is like a "book of accounts" but without names. Very unlike the traditional banking system, where the opening of a bank account calls for and requires a verified identity (KYC), a crypto wallet can be created and generated offline in seconds without any intermediary. This structural feature principally distorts the "third-party reporting" model that underpins our modern taxation. If we shed light on the OECD's traditional view, tax compliance relies on "information exchange between financial institutions and the state." When the "institution" is replaced by "code," and the "account" is a private key which is held on a USB drive, the system of information exchange breaks down.¹²

Moreover, if we talk about the development of privacy-enhancing technologies, it has actively weaponised this opacity. Various Tools such as "CoinJoins" or "Mixers" (e.g., Tornado Cash) complicates the transaction trail, making it computationally resource intensive to trace where the origin of funds is. Tools like CoinJoins or Mixers do not guarantee perfect anonymity against sophisticated forensic analysis. However, for tax authorities facing budget constraints, the cost of

¹² OECD, *Taxing Virtual Currencies: An Overview of Tax Treatments and Emerging Tax Policy Issues* (OECD 2020) 14; see also Hjalte Fejerskov Boas and Mona Barake, 'Enforcing Taxes on Cryptocurrencies' (2025) EU Tax Observatory Working Paper No 29.

de-anonymising these transactions often outweigh the recoverable revenue, creating a practical rather than theoretical difficulties in enforcement."

3.3 The AML-Tax Relationship: Deputising the Middleman

Considering the deficiency of a direct line of sight into the blockchain, tax authorities have increasingly relied on the alternative of the AML laws. The reasoning behind it is simple, *if tax authorities cannot track the asset, they must track the intermediary.*

3.3.1 Transfer of Funds Regulation: One of the most significant developments in this regard is the EU's "*Transfer of Funds Regulation* (Regulation (EU) 2023/1113), which became fully applicable on 30 December 2024". What this regulation does is that it implements the FATF "Travel Rule" into EU law, obligating that Crypto-Asset Service Providers must communicate the originator and beneficiary information alongside crypto-transfers.¹³

From a tax point of view, the TFR acts as a surveillance engine. It makes sure that every time user transfers funds from a regulated exchange (like Coinbase or Kraken) to another exchange, their identity will travel with the funds. Hence, tax authorities can subsequently request this data to reestablish a taxpayer's history. However, this system rests entirely on the presence of an intermediary who is compliant. It is effective for the "compliant majority" who use centralised services, but it is futile against those who operate exclusively on-chain.

3.3.2 The 2025 Data Gap: A crucial timing issue exists presently in the enforcement landscape. While the legal mandates for reporting are in place and in order one of which is the DAC8 and the OECD's CARF, the actual flow of data has lingered. As of recently, Member States are transposing DAC8, and the OECD expects the first automatic exchanges of CARF data to begin operating only in 2027.¹⁴

This creates a *blind spot* and a gap between 2024 till 2027. Sophisticated taxpayers are aware that while the laws have passed, but still the pipes for information exchange are not yet fully functional and operational. This gap period represents the peak of the Enforcement Gap, where the legal risk is excessive and up, but the probability of immediate detection remains minimal.

¹³ Regulation (EU) 2023/1113 of the European Parliament and of the Council of 31 May 2023 on information accompanying transfers of funds and certain crypto-assets [2023] OJ L 150/1 (Transfer of Funds Regulation).

¹⁴ OECD, 'Update on the implementation of the Crypto-Asset Reporting Framework' (OECD Tax Talks, December 2025). See also: *Statement on the timeline for the first exchanges under the CARF* (Global Forum on Transparency and Exchange of Information for Tax Purposes, November 2023).

3.4 The 'Unhosted' Blind Spot

Another significant challenge on tax enforcement is the "unhosted" (or self-hosted) wallet software or hardware which allows users to regulate their private keys without a third-party custodian.

Although EU has debated banning the “unhosted wallets” but it ultimately settled for a verification method. Under the TFR, CASPs must assess the risk of transfers to and from unhosted wallets and, for transactions exceeding €1,000, verify the identity of the owner on a risk-sensitive basis.¹⁵

However, there is a black hole lingering on for the tax authorities in the shape of unhosted wallets. If a taxpayer transfers Bitcoin from a regulated exchange to a Ledger Nano (hardware wallet) which in a unhosted wallet, the tax authority despite seeing the exit but it loses the journey. Let's suppose in its essence and practicality, the taxpayer might hold the asset (a non-taxable event in Germany after one year), then he swaps it for Stablecoins on a Decentralised Exchange (a taxable event in France) or even gift it to a child. None of these subsequent actions will be reported because they occur on-chain without an intermediary. This "DeFi Loophole" means that billions of euros in economic activity occur daily in a zone of zero tax visibility which will be surprising.

3.5 Cross-Border Jurisdictional Issues

To widen the enforcement gap, we can also see the inherent mismatch between the borderless technology and territorial tax sovereignty.

3.5.1 The Asset Confiscation Problem Collection is difficult even when a tax debt is created. In traditional enforcement process, a tax authority can issue a notice or an order to a bank to seize the funds. While in the case of crypto assets held in unhosted wallets, confiscation or seizure is possible but requires the use of the private key or device seizures. It is only possible if the taxpayer voluntarily surrenders the key or physical access to the device is obtained, the "taxable base" is statistically inaccessible. What this has done is that it has forced tax authorities to rely on "off-ramp" seizures waiting for the taxpayer to convert crypto back to fiat currency before intercepting the funds which seems way too inefficient, and anyone can easily counter doing that. Unlike bank accounts, unhosted wallets cannot be frozen by a central intermediary. While asset seizure is possible demonstrated by high-profile recoveries in the US and Germany involving compelled private key disclosure or device seizure it remains probabilistic and incomplete. It requires physical access to the taxpayer or their hardware, rendering remote enforcement significantly more difficult than in the traditional financial system.

3.5.2 Regulatory Arbitrage and 'Reverse Solicitation' The "Beyond" part of this thesis is suitable here. While the EU is tightening its grip, non-EU jurisdictions are offering a getaway. One of the major enforcement challenges is the "Reverse Solicitation" exemption in MiCA, which allows EU clients to originate service requests with non-EU firms. While the “ESMA has squeezed

¹⁵ Regulation (EU) 2023/1113, art 19a; see also European Banking Authority (EBA), ‘Guidelines on preventing the abuse of funds and certain crypto-assets transfers for money laundering and terrorist financing purposes’ (EBA/GL/2024/XX).

the guidelines on this practice,” it’s still present as a pathway for EU residents to access offshore, non-compliant exchanges that do not come under DAC8. Research and studies recently indicate that as domestic reporting is getting stricter, trading volumes shift to non-compliant offshore platforms, which very conveniently shifts the tax base to jurisdictions which have lower enforcement capacity and mechanisms.¹⁶

3.6 Outcomes & Talking Points

As the “Enforcement Gap” is more of a lack of reach rather than lack of law. With the help of TFR and AML directives, the EU has successfully enclosed the "intermediaries" of the crypto economy. That being said, the real aptitude and capacity of crypto peer-to-peer, permissionless finance remains structurally unaffected and defiant to territorial taxation. The more time the users can hold onto their own keys and continue trading on decentralized protocols, tax authorities will keep playing a game of catch-up. And will rely on their unpleasant tool of AML surveillance to enforce the precise rules of tax law. That is why enforcement gap can only be caught up with if the law makers correctly identify the gaps and keep up with the pace of rapid change.

¹⁶ Boas and Barake (n 12) finding that enforcement initiatives cause trading activity to shift to foreign platforms; see also ESMA, ‘Public Statement on the provision of crypto-asset services by third-country entities under MiCA’ (ESMA35-43-3137, 2025).

CHAPTER 4: The EU Response: MiCA and DAC8 “The Panopticon of the Blockchain”

4.1 The Two Towers of Compliance

The Two Towers or you can say twin pillars of the EU’s strategy for regulating and keeping a check on the digital asset economy is reliant on, the Markets in Crypto-Assets Regulation for practical supervision and the Eighth Directive on Administrative Cooperation for keeping tax transparency. Combined and seen together, these frameworks represent the most motivated attempt by any major jurisdiction to bring the "wild west" of crypto finance into the fold of the formal economy.

In this chapter critical evaluation is done on these two frameworks. The chapter delves into the argument on EU’s success on construing and coming up with mechanism to identify the crypto-assets through the MiCA while at the same time using DAC8 to report these assets. While the decentralized shadow economy remains blur both these systems are very effective, and you can say have created an “asymmetric transparency”. Authorities will soon possess a "Panopticon" view of compliant, centralized exchanges. Moreover, the chapter delves into the identification of a normative conflict between the EU’s drive for tax transparency and its fundamental commitments to data privacy, which I believe is going to create a constitutional tension in the upcoming years and is likely to define the next decade of litigation.

4.2 DAC8 “A Structure of Automatic Exchange”

DAC8 was officially adopted in October 2023 and is reaching its transposition deadline on 31 December 2025, “DAC8 (Directive (EU) 2023/2226)” principally changes the relationship between crypto-investors and the governments. It extends the automatic exchange of information regime which was previously only confined to bank accounts to crypto-assets, e-money, and Central Bank Digital Currencies.

4.2.1 Netting the 'Crypto-Asset Service Provider' so what DAC8 essentially does is that it creates a new reporting mandatory for "Reporting Crypto-Asset Service Providers". Very different to the Common Reporting Standards, which is used to target banks with proven compliance departments, what DAC8 does is, it specifically targets a diverse ecosystem of exchanges, brokers, and custodians.¹⁷ Three types of transactions are supposed to be reported as per this directive:

1. **Exchange Transactions:** Crypto-to-Fiat and, most importantly, Crypto-to-Crypto exchanges.
2. **Transfers:** Moving crypto currency from the exchange to an unhosted wallet (though the recipient of the unhosted wallet is often unidentified).

¹⁷ Council Directive (EU) 2023/2226 of 17 October 2023 amending Directive 2011/16/EU on administrative cooperation in the field of taxation [2023] OJ L 2023/2226 (DAC8), art 1(3) inserting art 8ad.

3. **Retail Payment Transactions:** Utilization of crypto to buy goods/services exceeding €50,000 (a high threshold that exempts most daily usage).

4.2.2 The 'Valuation Nightmare' for Business Through the lens of business law, the most critical requirement of DAC8 isn't the identification of the user but essentially the "valuation of the asset". Article 8ad, states that RCASPs are obliged to report the "fair market value" of the crypto asset during the time of transaction, denominated in fiat currency. Assets which are liquid like Bitcoin, this is very important. Although, for the long line of illiquid assets like "NFTs, governance tokens in small DAOs, or "memecoins" with thin order books determining a defensible "market value" is legally perilous". If a CASP reports a value of €1,000 based on a single erratic trade, and the tax authority assesses tax based on that figure, the CASP can consequently suffer liability risks from the client for "negligent reporting." Due to this EU exchanges are compelled to invest heavily in "pricing oracles" and algorithmic valuation tools, increasing the barriers to entry and fusing the market around a few dominant players.

4.3 Extraterritorial Deception

One of the key aspects and features of DAC8 is its aggressive extraterritoriality. It is not only applicable to EU-authorized CASPs but also to non-EU "Crypto-Asset Operators" who provide services to EU residents.

4.3.1 The 'Reverse Solicitation' Defense In order to get away from this web, non-EU firms have increasingly relied on the "Reverse Solicitation" exemption (Article 42 of MiCA). This doctrine states that "if a client exclusively initiates the service request, the non-EU firm does not need to comply with EU regulations". In this recent year, this exemption has become the primary battleground of enforcement. Offshore exchanges have structured their user interfaces to force European clients to click on the "I initiate this service" check box, creating a strong "paper shield" against DAC8. However, the ESMA has clarified that soliciting business is a factual test, not a contractual one. If an offshore firm runs ads in German or uses a *.fr domain*, the exemption is void. This has established a high-stakes game of "regulatory hide-and-seek" where non-EU firms try to service EU capital without trying to trigger EU reporting nexus.¹⁸

4.4 The Constitutional Conflict: DAC8 vs. GDPR

Analyzing DAC8 judiciously is unfulfilled without dealing with the "Privacy Paradox." The EU prides itself on the General Data Protection Regulations, and it should too, however what's interesting and surprising is that DAC8 mandates the mass collection and retention of sensitive financial data, often involving assets that reveal a user's entire transaction history (due to the public nature of the blockchain) for tax purposes.

¹⁸ European Securities and Markets Authority (ESMA), 'Guidelines on the conditions and criteria for the qualification of crypto-assets as financial instruments' (ESMA50-164-128, 2024). See also: Regulation (EU) 2023/1114 (MiCA), art 42 (Reverse Solicitation).

4.4.1 The 'Orde van Vlaamse Balies' Case Law Precedent In 2025, Law makers and scholars and creating similarities to the *Orde van Vlaamse Balies* judgment (C-694/20), where the CJEU put down parts of DAC6 (intermediary reporting) for violating on lawyer-client confidentiality.¹⁹ It is important to know that while CASPs are not lawyers, the principle of data proportionality remains. It is essential requirement of DAC8 to report of all crypto-to-crypto trades, arguably collecting vast amounts of data that are irrelevant for tax purposes (e.g., swapping one stablecoin for another with zero gain). In the case of businesses, this creates a double bind, staying compliant with DAC8 requires maximum data retention, while complying with GDPR requires data minimization. A CASP that reports too much data could face a GDPR class action and the one that reports too less faces DAC8 penalties. This normative conflict is likely to result in a CJEU challenge by privacy advocacy groups in the upcoming years. Legal scholars have raised concerns regarding the proportionality of this data collection. While the CJEU's ruling in *Orde van Vlaamse Balies* (C-694/20) limited reporting obligations based on professional privilege, Crypto-Asset Service Providers (CASPs) do not enjoy the same level of protected status as lawyers.

Nevertheless, a tension exists regarding the principle of data minimisation. DAC8 requires the bulk reporting of all crypto-to-crypto transactions, potentially accumulating vast datasets which is likely irrelevant to actual tax liability. While not inevitably fatal to the directive, this creates a plausible litigation risk, where the Court may eventually be asked to balance the state's interest in tax transparency against the fundamental rights of data subjects, particularly regarding the proportionality of retaining granular transaction histories.

4.5 MiCA "Practical Certainty & Fiscal Uncertainty"

Crucially the MiCA (Regulation (EU) 2023/1114) provides the definitional base for DAC8. However, the intersection between these two laws creates unintended friction.

4.5.1 The 'Grandfathering' Cliff Within recent times and lately, the "transitional period" for pre-existing CASPs (Article 143 MiCA) is expiring. The operating Firms under national regimes such as the Franch PSAN registration will not have to secure a full MiCA license. This establishes a holdup. Tax authorities rely on the list of authorised CASPs to know who to audit for DAC8 compliance. As thousands of small entities either fail to secure a license or they merge with larger entities, the "reporting population" is in flux. The "Grandfathering Cliff" means that 2026 will likely see a chaotic initial exchange of data, with many entities disappearing before they file their first DAC8 report.

4.5.2 Mismatch of the Definition Furthermore, it is pertinent to know that MiCA's definitions are designed for investor protection and not specifically taxation.

¹⁹ Case C-694/20 *Orde van Vlaamse Balies v Vlaamse Regering* [2022] ECLI:EU:C:2022:963. For the application to crypto, see: P. Valente, 'DAC8 and the GDPR: A collision course?' (2024) 64 *European Taxation* 12.

- For Example: MiCA exempts "unique and non-fungible" crypto assets such as NFTs from its scope "(Article 2(3))."
- The Gap: unlike the MiCA, DAC8 includes NFTs if they are used for "payment or investment." This gives rise to a situation where a CASP does not need a license to sell an NFT collection (under MiCA) but does need to report the buyers to the tax authority (under DAC8). This regulatory dissonance confuses businesses, who often assume that "unregulated" means "unreportable" and there is a huge difference between that.

4.6 Penalty Fragmentation and 'Forum Shopping'

Lastly, a major weakness that must be highlighted is in the final text of DAC8 was the failure to agree on a harmonised minimum penalty regime and this seems to be another problem in terms of enforcement. Moreover, the power to punish non-compliance was left to Member States.

4.6.1 The 'Soft Touch' Jurisdictions As Member States are nearing the final transposition in late 2025, some disagreements are visible.

- **The Hawks:** The Netherlands and France have come up with a proposal that penalties should be linked to a percentage of global turnover (similar to GDPR fines) for systemic non-reporting.
- **The Doves:** Smaller jurisdictions like Malta and Cyprus have chosen to go for the capped administrative fines. This encourages "Forum Shopping." A Crypto-Asset Operator (non-EU) looking to register in a "Member State of Identification" (as required by DAC8) is rationally incentivized to choose the state with the lowest penalties. This threatens to undermine the uniform application of the law, creating "entry points" and that is where enforcement issue occurs.²⁰

4.7 A House Built on Sand?

There is no denying the fact that DAC8 and MiCA represent a monumental legislative achievement. They have at least tried to replace the legal vacuum with a detailed compliance architecture. However, as this chapter has argued, the structure is way too fragile in my opinion. MiCA while being a harmonized regulatory layer when it interacts with non-harmonized layers creates a lot of tension. While its mandate doesn't include the scope of taxation it is still a crucial variant to discuss simultaneously with DAC8 as it has an indirect tax relevance. Hence coming back to the fragility of the structure, it rests on an unresolved tension between data privacy and tax transparency, relies on a "valuation" mechanism that is technically fraught, and is vulnerable to "forum shopping" due to penalty divergence. As 2026 approaches, the EU seems prepared to transition on a massive data-gathering machine. However, at the same time without a harmonised definition of what a taxable event is which was laid down in chapter 2 and without the ability to

²⁰ Comparison of transposition drafts: see *Dutch Ministry of Finance*, 'Implementation Act for Directive (EU) 2023/2226' (Explanatory Memorandum, July 2025) versus *Malta Commissioner for Revenue*, 'Guidelines on the implementation of DAC8' (Consultation Paper, November 2025). See generally: KPMG EU Tax Centre, 'DAC8: The Final Compromise' (Euro Tax Flash 512, 2023).

reach the unhosted and decentralized economy, this data may generate more litigation than revenue. The EU has built a robust fence like MiCA to regulate crypto assets, it still requires a similar framework which is as effective in the matters of regulating, streamlining and harmonizing taxonomy. While DAC8 can be termed as a “comprehensive camera”, it still needs a lot of work to ensure tax transparency.

CHAPTER 5: Beyond the EU: Global Standards, Strategic Divergence, and the Competitiveness of Tax Regimes

5.1 The Race for Regulatory Supremacy

The Crucial aspect that will be discussed in this chapter is to discuss global regulatory policies, while we can say that the European Union has constructed a fortress of compliance through the MiCA and the DAC8, the global digital asset economy does not operate in a vacuum. Capital, particularly in the digital realm, is super mobile. A "Crypto-Asset Service Provider" is not merely a technological entity but a legal fact that can relocate its domicile from Paris to London, Singapore, or Dubai very easily and if the regulatory friction becomes untenable.

This chapter unlike previous ones opens the analysis to the international dimension which is very important to examine the "Beyond" part of this thesis and also with respect to creating a uniformity in the globalised system on which organisations, such as the OECD, have been working on. Now since everything is not centralised anymore it is crucial to highlight these legal, regulatory and enforcement gaps from a globalised lens as well which would encourage future policy making in that direction. The chapter thus contrasts the EU's codified and civil law approach with the evolving frameworks of the OECD, the United Kingdom, the United States, and emerging "crypto hubs". It argues that as of late 2025, a "Great Divergence" is emerging. "While the OECD attempts to standardize reporting (the "what"), jurisdictions are competing fiercely on definition (the "what is it")." This competition is no longer a race to the bottom (deregulation) but a race to "quality" where multinational businesses seek the jurisdiction that offers the optimal balance of *tax* transparency and private law certainty.

5.2 The OECD and CARF "Structure of Global Surveillance"

The OECD has historically driven global tax transparency through the Common Reporting Standard (CRS) and one cannot deny it has been very effective. However, the CRS was designed for the legacy banking system, predicated on the existence of "Financial Institutions" holding "Financial accounts." With the growing influence of crypto assets, which can be held very easily in unhosted wallets without an intermediary, rendered the CRS framework structurally obsolete and impractical.

5.2.1 From CRS to CARF: To plug this gap, the OECD tried and introduced the "Crypto-Asset Reporting Framework (CARF)." Unlike the CRS, which looks for "accounts," the CARF looks for "transactions." What it means is it requires that "Reporting Crypto-Asset Service Providers" (RCASPs) collect data on all transfers of "Relevant Crypto-Assets" and exchange this information automatically with tax authorities.²¹

²¹ OECD, *Crypto-Asset Reporting Framework and Amendments to the Common Reporting Standard* (OECD Publishing, 2023) 14.

The depth of the CARF's structure lies in its expansive definition of "intermediary." While the EU's DAC8 aligns with this, the global challenge is the "Nexus Rule." Under the CARF it mandates that reporting of a company shall be done both where it is resident taxpayer and also where it is operating in order to prevent "reverse solicitation" loophole. However, the effectiveness of CARF depends entirely on the "match rate" the ability of a receiving tax authority to match a wallet address to a taxpayer ID. Within recent times, pilot programs suggest this match rate remains below 40% for non-custodial transfers, highlighting a persistent technical gap in the global standard.²²

5.2.2 The 2026/2027 Fracture: The accomplishment of any Automatic Exchange of Information (AEOI) regime depends on reciprocity. For example, if Country A reports to Country B, but Country B does not report to Country A, the system will falter. A critical timeline fracture has emerged in the global implementation of CARF:

- **The Euro-Atlantic Bloc (Early Adopters):** The EU via DAC8 and the UK through the Reporting Crypto asset Service Providers Regulations 2025 have synchronised their start dates for 1 January 2026. This creates a unified "transparency zone" across Europe which is a positive sign.
- **The American Lag:** Despite the euro transparency zone the United States, despite being a primary architect of the OECD framework, has delayed its domestic implementation. The IRS final regulations on "Broker" reporting (T.D. 10021), was launched in the late 2024 due to which it has postponed the effective date for certain cost-basis reporting requirements until 2027.²³

This asymmetry creates a "regulatory arbitrage window" throughout 2026. A global crypto exchange operating in both London and New York faces a separate reality, it must report every transaction of its UK clients to HMRC immediately and in effect, while its US clients remain in a transitional limbo. The uneven and inconsistent playing field further benefits the big traders to shift their liquidity to US domiciled desks during this period undermining the "level playing field" that CARF sought to establish.

5.3 The 'Third Category' Advantage in the United Kingdom

While the EU has focused its eyes on its public law regulation MiCA / DAC8, the UK has executed a strategic pivot by laying importance on private law certainty. This distinction is crucial in terms

²² Global Forum on Transparency and Exchange of Information for Tax Purposes, *Peer Review of the Automatic Exchange of Financial Account Information 2025* (OECD 2025) para 88. See also: Chainalysis, *the 2025 Geography of Cryptocurrency Report* (Chainalysis 2025), noting the difficulty in identifying beneficial owners of unhosted wallets.

²³ US Department of the Treasury (IRS), 'Gross Proceeds and Basis Reporting by Brokers and Determination of Amount Realized and Basis for Digital Asset Transactions' (TD 10021, 2024) 89 FR 10000. See also: Grant Thornton, 'IRS Finalizes Crypto Broker Rules with Delayed Implementation' (Tax Alert, January 2025).

of International Business Law, businesses care about regulation, but they care even more about property rights.

5.3.1 The Property (Digital Assets etc) Act 2025 On 2nd of December 2025, the UK implemented the “Property (Digital Assets etc) Act 2025”, executing the Law Commission’s recommendation to explicitly recognise a "third category" of personal property. Traditionally, English law has only been recognizing the "things in possession" (tangible objects) and "things in action" (rights claimable by lawsuit, like shares). Crypto assets fit neither category perfectly, they are not tangible, yet they are transferable without a lawsuit.²⁴

By statutorily creating this third category, we can say that the UK has solved the "insolvency risk" that plagues and created issues for the EU markets. In the EU, if a crypto exchange goes bankrupt, the question of whether client assets are part of the insolvency estate often depends on vague national interpretations of "ownership." In the UK, the Property Act 2025 is very streamlined, and it provides liquidators and investors with a clear statutory basis to ring-fence and protect these assets. For institutional investors for example pension funds looking to allocate to Bitcoin ETFs, this private law certainty reduces counterparty risk significantly more than MiCA’s prudential rules.

5.3.2 The 'Badges of Trade' vs. Rigid Codification Moreover what makes UK even more practical is that it is driven by flexibility such as the UK tax authority (HMRC) has maintained a flexible, case-law-driven approach to classification which is unlike EU and its civil law structure for example in Germany’s rigid "1-year speculation period" or France’s "flat tax," the UK applies the "Badges of Trade" test to distinguish between investment activity (Capital Gains Tax) and trading activity (Income Tax). This Common Law flexibility allows the UK to adapt to new DeFi innovations (like "liquid staking" or "restaking") without waiting for new legislation and then subsequently its enforcement. If a new token behaves like a security, the courts can instantly treat it as such. This "regulation by analogy" is proving more adaptable to the speed of crypto innovation than the EU’s "regulation by code."

5.4 The United States

In my opinion The United States is far behind both the EU and UK and it has the most complex and litigious environment for digital assets. It also lacks frameworks like "MiCA-style" framework, the US approach has been characterised by "regulation by enforcement" and intense statutory interpretation issues.

5.4.1 The 'Broker' Controversy and the “Infrastructure Act” A major pertinent issue and challenge in US taxation lies in its definition introduced by the “Infrastructure Investment and Jobs Act (2021).” The Act has expanded the definition of a "Broker" under “Section 6045 of the Internal

²⁴ *Property (Digital Assets etc) Act 2025*, s 1(1). This act codified the recommendations found in: Law Commission, *Digital Assets: Final Report* (Law Com No 412, 2023).

Revenue Code” to include “any person who (for consideration) is responsible for regularly providing any service effectuating transfers of digital assets on behalf of another person.”²⁵

This is a definition which is broad enough to theoretically capture non-custodial actors, such as “wallet developers, miners, or DeFi protocol governance token holders.” During 2024 the US treasury was challenged by the struggle to narrow down this definition in its final regulations to avoid capturing entities that technically cannot comply (because they do not hold customer KYC data). The result, which was finalised in late 2024, was a compromise that exempted pure software developers but retained strict reporting for “DeFi interfaces” that maintain a front-end website. This has led to a constitutional challenge under the “First Amendment (code as speech) and the Fourth Amendment (unreasonable search and seizure), creating a climate of immense legal uncertainty that is completely different with the clarity of the EU’s DAC8.”

5.4.2 The 'Major Questions' Doctrine A different and important factor of the US law is its Supreme Court’s “Major Questions Doctrine”, which limits the ability of agencies (like the SEC or IRS) to regulate economically significant areas without explicit Congressional authorisation. In 2025, major crypto exchanges (e.g., Coinbase, Kraken) have taken the benefit of this doctrine to challenge the SEC’s assertion of jurisdiction. They have debated that if Congress wanted the SEC to regulate crypto as securities, it would have passed the Digital Asset Market Clarity Act (which stalled in the Senate) and in my opinion it is a pertinent point. This “separation of powers” gridlock means that while the US is the largest market by capital, it is the most legally volatile and vulnerable at the same time. For a multinational business, the US offers depth of liquidity, but it requires a high-risk tolerance for regulatory litigation.

5.5 Singapore and the UAE “The Third Pillars”

To completely understand the “And Beyond” scope, one must acknowledge the rise of “Crypto-Hubs” that compete directly with the EU and US. Major Jurisdictions like Singapore and the United Arab Emirates (specifically Dubai) have emerged as a “Third Pole” in the global order and are very essential to our analysis.

5.5.1 The Specialist Regulator Model Unlike the US (fragmented between SEC/CFTC) or the EU (fragmented between 27 National Competent Authorities), Dubai has successfully established the Virtual Assets Regulatory Authority (VARA) as a dedicated, single-sector regulator. From a tax point of view, these jurisdictions offer a compelling proposition, zero or very low direct tax on capital gains, combined with high regulatory standards to avoid being blacklisted by the FATF.

²⁵ Infrastructure Investment and Jobs Act 2021, Pub L No 117-58, § 80603, 135 Stat 429 (codified at 26 USC § 6045(c)(1)(D)). See also: *Coin Center v. Yellen*, No. 22-cv-2037 (N.D. Fla. 2024) (challenging the constitutionality of the reporting requirements).

- **Singapore:** The Monetary Authority of Singapore (MAS) has shifted towards a rigid consumer protection stance (banning retail speculation) while encouraging institutional implementation.
- **UAE:** is offering "tax neutrality" within Free Zones.

The relevance for the EU is critical in this matter as businesses are effectively bifurcating their operations. They create their "compliance HQ" in the EU (to access the MiCA passport) but locate their "trading engines" and intellectual property in the UAE or Singapore to optimise tax efficiency and can get away with it easily. This phenomenon, known as "protocol partitioning," undermines the EU's tax base even as MiCA succeeds in regulating market conduct and one must say that these gaps while still present in a centralized structure where base erosion occurs and OECD has been working on its uniform enforcement globally a new challenge has already arisen in form of these crypto-assets where base erosion and profit shifting can occur way too easily and without a need of complicated transfer of assets from one jurisdiction to another.

5.6 Legal Certainty as a Competitive Asset & Analysis

The comparison across these jurisdictions reveals that "tax competition" has evolved and is now posing more challenges than ever before. It is no longer just about rates (e.g., 0% vs 30%). It is about computational certainty.

- **The EU Model:** Offers a "Compliance Garden." It is walled, strict, and costly, but safe. You know exactly what an "E-Money Token" is.
- **The UK Model:** Is Offering "Property Certainty." It is the best jurisdiction for holding assets in trust or bankruptcy-remote vehicles due to its flexibility and adaptability.
- **The US Model:** Offers "Market Depth" but "Legal chaos and complexity." It is indispensable for liquidity but toxic for compliance planning.

For an international business law perspective, the optimal strategy in years to come is often a hybrid structure, issuing the token in a Common Law jurisdiction (UK/BVI) to secure property rights, obtaining a MiCA license in a streamlined EU state for example in France for distribution, and hedging regulatory risk by avoiding US nexus where possible until it is developed and streamlined.

5.7 Outcomes

The international landscape of the upcoming years is defined by a tight race between tax cooperation (CARF) and legal competition. The EU and UK have successfully locked step on the mechanics of reporting, effectively limiting the web on simple evasion. However, they are competing fiercely on legal design which is hard to uniform and streamline easily. The UK's move to codify digital property rights creates a significant "moat" around its financial services sector, challenging the EU's thought that MiCA alone would secure global dominance. Meanwhile, still the US remains behind the race and is cautionary in a tale of what happens when tax enforcement gets ahead of legal definition. As the next chapter will conclude, the only way to resolve this global fragmentation is not more regulation, but a much more enhanced optimisation of the underlying legal concepts of property and income. Moreover, Public International Law plays a huge role in this cause where OECD will play a huge role in optimising a clear structure towards future harmonization of global tax laws.

CHAPTER 6: Conclusion and Recommendations: Towards a Harmonised Tax Taxonomy

6.1 The Unresolved Dichotomy

This thesis has tried to investigate the challenges of defining and regulating digital assets within the European Union and beyond, specifically through the lens of tax law and international business compliance. Although the attempt is to cover the foundational loopholes, there still remains a lot to cover in terms of future policy making, legislation, enforcement and the role of public international law and treaties. The central finding of this research has been regarding the fact that fundamental dichotomy exists at the heart of the modern digital economy, the divergence between financial regulation, which has become increasingly harmonised (MiCA), and taxation, which remains fragmented, territorially bound, and legally unclear.

As of this date, the EU has successfully erected a surveillance infrastructure through DAC8 and the “Transfer of Funds Regulation.” However, as this thesis has raised concerns, surveillance is not a substitute for definition. Knowing that a transaction occurred is useless if the legal system cannot agree on what that transaction is whether it is a “currency exchange, a barter of commodities, or a realisation of an intangible asset.” This chapter further synthesizes and concludes the findings of the preceding analysis, explicitly contrasting the judicial approaches of the CJEU and Common Law courts, and tries to propose a principled framework for future policy.

6.2 Findings & The Three Major Grey Areas

In the research questions posed in Chapter 1 above revealed three distinct "gaps" that currently plague the taxation of crypto assets:

6.2.1 The Classification Gap Without a doubt the lack of a uniform definition has created a legal environment where the same asset changes its nature as it crosses a border especially in the EU member states. As analysed in Chapter 2, the EU is trapped by the legacy of “*Skatteverket v Hedqvist*.”²⁶ By laying down the definition of Bitcoin as a “means of payment” for VAT purposes, the Court of Justice of the European Union (CJEU) correctly identified the function of the asset but didn’t touch upon its substance for direct tax purposes as direct tax is left under the sovereignty of member states which leads to disharmony. This stands a deep contrast to the emerging “Property” consensus in Common Law jurisdictions which is discussed below, creating a friction that MiCA’s regulatory definitions have failed to resolve.

6.2.2 Major Enforcement Gap The analysis in Chapter 3 demonstrated which touched upon the second research area is that enforcement is currently reliant on a “chokepoint” strategy regulating the intermediaries (CASPs) but not the beneficiaries or the one who initiates the transaction. However, the rise of “unhosted” wallets and DeFi fundamentally challenges this model and opens wide gaps. The reliance on AML data to enforce tax obligations is a necessary step for the time

²⁶ Case C-264/14 *Skatteverket v David Hedqvist* [2015] ECLI:EU:C:2015:718.

being but it is an insufficient stopgap. It can cater to the leakage by reducing it marginally but is ineffective when it comes to resolving it. It effectively creates a two-step system, a highly transparent "white market" for compliant institutional investors, and a dark, unenforceable "grey market" for peer-to-peer actors.

6.2.3 The Regulatory Gap Chapter 4 and 5 have touched upon and concluded that while DAC8 and CARF provide the mechanism for data exchange, they do not harmonise the tax base. The research highlighted that the EU's focus on prescriptive legislation (Civil Law) is currently losing the "legal certainty" race to the UK's flexible property law approach (Common Law). The delay in US implementation of CARF further fragments the global landscape, incentivizing regulatory arbitrage. The key to this also rests upon but is not limited to Public International Law, tax treaty optimization and would need considerable efforts from OECD to limit base erosion and profit shifting while collaborating with the states.

6.3 Case Law Analysis & Judicial Divergence

To understand the depth of the "Classification Gap," it is necessary to contrast how courts in different legal traditions have attempted to fill the void left by legislators. The judiciary has become the de facto regulator of tax definitions.

6.3.1 The CJEU's Approach In the EU, the jurisprudence remains anchored to Hedqvist till now. While this judgment was pivotal for VAT neutrality and helped in that regard, it has didn't offer any help with regards to Direct Taxation. For example, lets suppose if a German investor exchanges Bitcoin for Ethereum, the German Federal Fiscal Court (Bundesfinanzhof) has had to wrestle with whether this is a "sale of a private economic good" (Section 23 EStG). In a significant 2023 ruling (*BFH IX R 3/22*), the German court confirmed that "crypto-assets are economic goods capable of generating taxable income", effectively rejecting the argument that they are mere "computer code" without value.²⁷ This ruling was critical for business certainty in Germany, but it highlights the disproportionality that the German court's definition applies only in Germany and not in any other Member State. There is no pan-European "Supreme Court of Direct Taxation," leaving the definition of the asset class to fracture across 27 national supreme courts.

6.3.2 The Common Law's Flexible Approach In contrast, Common Law courts have been more aggressive in defining the proprietary nature of crypto assets, which underpins their tax treatment.

- **Property Status:** The critical case of *AA v Persons Unknown* [2019] in the UK High Court (and reaffirmed in subsequent crypto-fraud cases) established that crypto-assets are "property" capable of being held on trust as well.²⁸ This judicial certainty allows UK tax

²⁷ Bundesfinanzhof (BFH), Judgment of 14 February 2023, IX R 3/22. The court confirmed that cryptocurrencies constitute 'other assets' within the meaning of s 23(1) sentence 1 no 2 of the Income Tax Act (EStG).

²⁸ *AA v Persons Unknown* [2019] EWHC 3556 (Comm). See also the New Zealand landmark case *Ruscoe v Cryptopia Ltd (in Liquidation)* [2020] NZHC 728, which provided the rigorous 'four-limb' test for defining digital assets as property, widely cited in Common Law jurisdictions.

law (HMRC) to seamlessly apply existing Capital Gains Tax rules without needing new primary legislation and builds an ecosystem of adaptability.

- **Trading vs. Investing:** The UK courts have also provided more clarity on when activity becomes a taxable "trade." In *Chekin v HMRC* [2020] (regarding share trading, but applied by analogy to crypto), the tribunal clarified the high threshold for "badges of trade."²⁹ This case law effectively "protects most retail crypto investors from high Income Tax rates, categorizing their activity as Capital Gains."
- **The Staking Controversy:** In the US, the case of *Jarrett v. United States* (2022) exposed the fragility of tax definitions and exposed many loopholes. The taxpayers argued that "staking" rewards (new tokens created by the protocol) were akin to a baker baking a loaf of bread and thus a new property created which does not mean that income is realized. While the IRS offered a refund to resolve the case, the precedent remained undecided and this shows the principal flaw in the system when it treats new digital asset creation as an income.³⁰

6.3.3 The Implications for Businesses For a cross-border business, this judicial divergence is costly. A "staking" service provider operating in the EU must account for the fact that Jarrett-style arguments might work in Common Law jurisdictions (UK/US) but fail in codified Civil Law jurisdictions such as the Germany or France, where statutory definitions are more rigid and strict.

6.4 Principled Criteria for Future Policy

Based on the analysis done in this research regarding MiCA, DAC8, and the comparative case law, this thesis proposes three normative recommendations to resolve the current deadlock and future policy making.

Recommendation A: Streamlining and Harmonising MiCA & DAC8 Having critically analysed the MiCA and DAC8 it feels like the EU needs to create uniformity among its regulations and frameworks. DAC8 alone. Currently, MiCA acts as a regulatory gatekeeper, while DAC8 acts as a tax reporting tool, yet they often operate in silos. This disconnect is a primary cause of the "Definitional Quagmire" identified in this thesis.

- **Proposal:** To resolve this, the EU should mandate that the definitions established in MiCA serve as the binding classification for tax reporting under DAC8. Put simply, if an asset is defined as an "E-Money Token" for regulatory purposes under MiCA, it should automatically be treated as a currency-equivalent for tax reporting across all Member States.

²⁹ *Chekin v Commissioners for HM Revenue and Customs* [2020] UKFTT 449 (TC). While regarding share trading, this case is the leading authority cited by HMRC to distinguish between a 'hobby' investor (Capital Gains) and a 'trader' (Income Tax).

³⁰ *Jarrett v. United States*, No. 3:21-cv-00419 (M.D. Tenn. 2022). See also: Joshua Jarrett, 'Complaint for Refund of Taxes' (2021), arguing that staking rewards are created property (like crops), not realised income. Although the case was settled, the legal argument remains the primary challenge to US IRS definitions.

- Justification & why it will be effective: By harmonising these two frameworks, the EU would eliminate the ambiguity where one asset has multiple legal identities. This uniformity would significantly reduce compliance difficulties for businesses, allowing them to build a single reporting system that satisfies both regulatory and tax obligations.

Recommendation B: 'Compliance by Design' (The Technological Solution) The "Enforcement Gap" in Chapter 3 cannot be closed by chasing unhosted wallets and it doesn't seem to be a wise strategy. It must be closed by integrating tax into the protocol.

- Proposal: Regulatory policy should incentivize the development of "Tax-Wrapper" smart contracts. These are optional protocols that automatically segregate a portion of transaction fees into a "tax vault" attached to the user's identity.
- Justification: As argued in the analysis of DeFi, retrospective reporting is computationally impossible for high-frequency on-chain activity, and one cannot continue running behind it forever and tracing it. Real-time, algorithmic tax withholding is the only resolute and better long-term solution for the unhosted economy.

Recommendation C: Global Streamlining of 'Realisation Events' The greatest barrier to international business is the disagreement on when a tax event occurs or arises.

- Proposal: The OECD should lead an agreement and acceptance on "Crypto-to-Crypto" neutrality.
- Justification: Taxing an interchange between currencies like from Bitcoin to Ethereum is administratively tough, time taking and liquidity-draining. Moving to a "Cash-Out" taxation model in which there is taxing only when crypto is converted to fiat or goods would drastically simplify compliance and will significantly reduce the reporting burden under CARF and align tax law with the economic reality that crypto-to-crypto trades are often merely portfolio rebalancing, not profit-taking.

6.5 Closing Comments

It's amazing how the path of crypto assets from the fringes of the internet to the centre of the global financial system is now almost complete. The legal system, however, is still catching up as it does most of the times. Since this date, the European Union stands as a global leader in prudential regulation (MiCA) and tax transparency (DAC8) despite all the critique done above. However, one cannot disagree that it remains enveloped in fiscal certainty.

The "quagmire" identified in this thesis is not merely a technical nuisance, it is a structural flaw in the Internal Market. By allowing tax definitions to drift apart while harmonising financial rules, the EU has created a single market for trading of crypto, but a very weak market for taxing it. Until and unless this disproportionality is resolved either through a "unified tax taxonomy" or even through judicial mergence the promise of a frictionless digital asset economy will remain unfulfilled. The law must move forward from simply watching the blockchain as part of surveillance to understanding it and defining it properly. Only then can the balance between innovation and enforcement be restored. Till then the global world is a playground for the traders who can enjoy while the Public International Law, legal, regulatory and enforcement structures are developing and trying to catch up in this cat and mouse game and who knows till the time they catch up the world might've entered a completely different era.

Bibliography

Table of Cases

European Union

- Case C-264/14 *Skatteverket v David Hedqvist* [2015] ECLI:EU:C:2015:718
- Case C-694/20 *Orde van Vlaamse Balies v Vlaamse Regering* [2022] ECLI:EU:C:2022:963

United Kingdom

- *AA v Persons Unknown* [2019] EWHC 3556 (Comm)
- *Chekin v Commissioners for HM Revenue and Customs* [2020] UKFTT 449 (TC)
- *Ruscoe v Cryptopia Ltd (in Liquidation)* [2020] NZHC 728 (New Zealand High Court, cited in UK)

United States

- *Coin Center v. Yellen*, No. 22-cv-2037 (N.D. Fla. 2024)
- *Jarrett v. United States*, No. 3:21-cv-00419 (M.D. Tenn. 2022)

Germany

- Bundesfinanzhof (BFH), Judgment of 14 February 2023, IX R 3/22

Table of Legislation

European Union

- Council Directive (EU) 2011/16/EU on administrative cooperation in the field of taxation [2011] OJ L 64/1
- Council Directive (EU) 2023/2226 of 17 October 2023 amending Directive 2011/16/EU on administrative cooperation in the field of taxation [2023] OJ L 2023/2226 (DAC8)
- Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data (GDPR) [2016] OJ L 119/1
- Regulation (EU) 2023/1113 of the European Parliament and of the Council of 31 May 2023 on information accompanying transfers of funds and certain crypto-assets [2023] OJ L 150/1 (Transfer of Funds Regulation)

- Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto assets (MiCA) [2023] OJ L 150/40

United Kingdom

- Property (Digital Assets etc) Act 2025
- The Reporting Cryptoasset Service Providers (Due Diligence and Reporting Requirements) Regulations 2025, SI 2025/744

United States

- Digital Asset Market Clarity Act of 2025, HR 3633, 119th Cong (2025)
- Infrastructure Investment and Jobs Act 2021, Pub L No 117-58, 135 Stat 429
- Internal Revenue Code, 26 USC § 6045

National Legislation (Selected)

- Code général des impôts (General Tax Code), art 150 VH bis (France)
- Einkommensteuergesetz (Income Tax Act), s 23 (Germany)
- Loi n° 2019-486 du 22 mai 2019 relative à la croissance et à la transformation des entreprises (PACTE Law) (France)

Secondary Sources

Books

- De Filippi P and Wright A, *Blockchain and the Law: The Rule of Code* (Harvard University Press 2018)

Journal Articles

- Ceron P and Yanev A, ‘The intersection of AML and Tax Evasion in the Crypto sphere’ (2022) 30(2) *Journal of Financial Regulation and Compliance* 155
- Cockfield A and McGill S, ‘Shaping International Tax Law and Policy in the Digital Age’ (2018) 18 *Wisconsin International Law Journal*
- Low K and Teo E, ‘Bitcoins and Other Cryptocurrencies as Property?’ (2017) 9(2) *Law Quarterly Review* 235
- Marian O, ‘Are Cryptocurrencies Super Tax Havens?’ (2013) 112 *Michigan Law Review First Impressions* 38
- Valente P, ‘DAC8 and the GDPR: A collision course?’ (2024) 64 *European Taxation* 12

Official Reports, Guidelines & Policy Documents

- Bundesministerium der Finanzen (BMF), ‘Einzelfragen zur ertragsteuerrechtlichen Behandlung von virtuellen Währungen und von sonstigen Token’ (10 May 2022) IV C 1 - S 2256/19/10003 :001
- Chainalysis, *The 2025 Geography of Cryptocurrency Report* (Chainalysis 2025)
- European Banking Authority (EBA), ‘Final Report on Guidelines on the delineation between different categories of crypto assets’ (EBA/GL/2024/XX)
- European Securities and Markets Authority (ESMA), ‘Guidelines on the conditions and criteria for the qualification of crypto-assets as financial instruments’ (ESMA50-164-128, 2024)
- Financial Action Task Force (FATF), *Updated Guidance for a Risk-Based Approach to Virtual Assets and Virtual Asset Service Providers* (FATF 2021)
- Global Forum on Transparency and Exchange of Information for Tax Purposes, *Peer Review of the Automatic Exchange of Financial Account Information 2025* (OECD 2025)
- HM Revenue & Customs (HMRC), ‘Cryptoassets Manual’ (HMRC, updated November 2025)
- Internal Revenue Service (IRS), ‘Notice 2014-21: Virtual Currency Guidance’ (2014) IRB 2014-16
- Internal Revenue Service (IRS), ‘Revenue Ruling 2019-24: Virtual Currency Hard Forks and Airdrops’ (2019)
- KPMG EU Tax Centre, ‘DAC8: The Final Compromise’ (Euro Tax Flash 512, 2023)
- Law Commission, *Digital Assets: Final Report* (Law Com No 412, 2023)
- OECD, *Crypto-Asset Reporting Framework and Amendments to the Common Reporting Standard* (OECD Publishing 2023)
- OECD, *Taxing Virtual Currencies: An Overview of Tax Treatments and Emerging Tax Policy Issues* (OECD 2020)
- US Department of the Treasury (IRS), ‘Gross Proceeds and Basis Reporting by Brokers and Determination of Amount Realized and Basis for Digital Asset Transactions’ (TD 10021, 2024) 89 FR 10000